NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT

B. TECH/AIML,CSE,CST,EIE,CE,ECE,EE,IT,ME/EVEN/2ND SEM/R21/M201/2022-2023
YEAR: 2023

MATHEMATICS-II
M201

TIME ALLOTTED: 3 HOURS FULL MARKS: 70

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable

GROUP – A
(Multiple Choice Type Questions)

1. Answer any ***ten*** from the following, choosing the correct alternative of each question: **10×1=10**

SL	Question	Marks	Co	Blooms Taxonomy Level
(i)	Geometrically the Lagrange's interpolation formula for two points of interpolation represents a a) Parabola b) Straight line c) Circle d) None	1	3	Apply
(ii)	The degree of precision of Trapezoidal rule is a) 1 b) 2 c) 3 d) 5	1	2	Understand
(iii)	$L^{-1}\left(\frac{5}{s^2-a^2}\right) =$ a)sin at b)sinh at c)cos at d)cosh at	1	1	Recall
(iv)	R K method has a truncation error of the order of a) h b) h^2 c) h^4 d) h^5	1	5	Evaluate
(v)	In Simpson's one third rule the error will be zero, if $f(x)$ be a polynomial of degree a)greater than three b) less than equal to three c) greater than four d) four	1	3	Apply

(vi)	$\frac{1}{D-3}e^{3x}$ is equal to a) xe^{3x} b) e^{3x} c) $3e^{3x}$ d) none	1	4	Analyze
(vii)	The general solution of the ordinary differential equation $\frac{d^2y}{dx^2} + 4y = 0$ a) $Ae^{2x} + be^{-2x}$ b) Acos2x+Bsin2x c) (A+Bx)e^{2x} d) None of These	1	5	Evaluate
(viii)	The particular integral of $\frac{d^2y}{dx^2}$ + y = cos x is (a) $\frac{1}{2}$sin x (b) $\frac{1}{2}$cos x (c) $\frac{1}{2}$xsin x (d) xsin x	1	2	Understand
(ix)	The number of significant digits in 0.00340 is a) 3 b) 4 c) 5 d) 6	1	3	Apply
(x)	Which of the following is true? a) $L^{-1}\{f(s-a)\} = e^{-at}L^{-1}\{f(s)\}$ b) $L^{-1}\{f(s-a)\} = L^{-1}\{f(s)\}$ c) $L^{-1}\{f(s-a)\} = e^{at}L^{-1}\{f(s)\}$ d) $L^{-1}\{f(s+a)\} = e^{at}L^{-1}\{f(s)\}$	1	1	Recall
(xi)	The integrating factor of the differential equation is $\frac{dy}{dx}$ +5y=2x a) e^{5x} b) e^{5y} c) e^{2x} d) none of these	1	4	Analyze
(xii)	The equation $(3x^2 + py)dx + (-6y^2 + qx)dy = 0$ is exact if a) P+q=0 b) p-q=0 c) 3p+q =0	1	5	Evaluate

d) p≠q

GROUP – B
(Short Answer Type Questions)
(Answer any three of the following) 3 x 5 = 15

SL	Question	Marks	Co	Blooms Taxonomy Level
2.	Find the missing value in the following table: (see table below)	5	3	Apply
3.	Solve the differential equation $\frac{dx}{dy} + \frac{x}{ylogy} = \frac{1}{y}$	5	5	Evaluate
4.	Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's 1/3rd rule taking six sub-intervals, correct up to three decimal places.	5	2	Understand
5.	Evaluate $\int_0^\infty \frac{e^{-3t}-e^{-6t}}{t} dt$	5	1	Recall
6.	Find inverse laplace transform of $\frac{8e^{-3s}}{s^2+4}$	5	4	Analyze

Table for Q2:

x :	2	4	6	8	10
y :	5.6	8.6	13.9	__	35.6

GROUP – C
(Long Answer Type Questions)
(Answer any three of the following) 3 x 15 = 45

SL		Question	Marks	Co	Blooms Taxonomy Level
7.	(i)	Apply Lagrange's interpolation formula to find $f(x)$, if $f(1) = 2, f(2) = 4, f(3) = 8$, and $f(4) = 16$.	7	2	Understand
	(ii)	Solve the following simultaneous equations: $\frac{dx}{dt}$ - 7x + y = 0, $\frac{dy}{dt}$ - 2x + 5y = 0	8	1	Recall
8.	(i)	Find the general and singular solution of py=$p^2(x-b)+a$ where p=$\frac{dy}{dx}$	7	5	Evaluate
	(ii)	Apply convolution theorem of Laplace transformation to	8	3	Apply

		evaluate $L^{-1}\{\frac{p}{(p^2+a^2)^2}\}$			
9.	(i)	Apply the method of variation of parameters to solve $$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 9e^x$$	7	2	Understand
	(ii)	Using Weddle's rule, evaluate $\int_0^3 x dx$, taking 6 equal sub-intervals.	8	1	Recall
10.	(i)	Given that, $\frac{dy}{dx} = xy + y^2, \quad y(1) = 1$ Use Runge-Kutta method of fourth order to find $y(1.1)$, correct to three decimal places.	8	3	Apply
	(ii)	Solve the following differential equation: $(x^2 D^2+4xD+4)\ y = \log x$	7	5	Evaluate
11.	(i)	Find , from definition , the Laplace Transform of the function f(t) defined by $f(t) = 0,\ 0 < t \leq 1$; $= t,\ 1 < t \leq 2$; $= 0,\ t > 2$	8	2	Understand
	(ii)	Solve: $y=px + \sqrt{p^2+1}$ where $p=\frac{dy}{dx}$	7	1	Recall